TED (21) – REVISION 2021

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

Model Question Paper I

SIGNALS AND SYSTEMS

Time: 3 Hour *Maximum Marks: 75*

PART A

I. Answer **all** questions in one word or one sentence. Each question carries 1 mark

(9*1=9 marks)

Module outcome Cognitive level

1	Define signals.	M 1.01	R
2	Write any two applications of signals and systems.	M 1.01	R
3	Draw a discrete time signal.	M 2.02	U
4	Define a dynamic system.	M 2.01	R
5	What is a time invariant system?	M 2.03	U
6	Convolution in time domain corresponds to …………..in frequency domain.	M3.02	U
7	Define Fourier series.	M 3.01	U
8	What is ROC?	M 4.02	U
9	Write the Laplace transform for unit impulse.	M 4.01	U

PART B

II. Answer any **eight** questions from the following, each question carries 3 marks.

(8 * 3=24 marks)

Module outcome Cognitive level

1	Differentiate even and odd signals.	M 1.03	U
2	Illustrate addition of signals with figures.	M 1.04	U

3	Write notes on any two basic signals used in communication.	M1.02	U
4	Differentiate linear and non-linear system.	M 2.03	U
5	Define a continuous time system with example.	M 2.02	U
6	What are the Fourier representation of four classes of signals?	M 3.01	R
7	Define sampling theorem.	M 3.03	R
8	Write the time shifting property of Fourier series.	M 3.02	R
9	Find the Laplace transform of ramp function.	M 4.01	U
10	Find the inverse Laplace transform of $\frac{1}{s}$	M4.04	U

PART C

III. Answer ALL questions. Each question carries 7 marks.

(6*7=42 marks)

Module outcome Cognitive level

1	Explain with figure any four elementary signals.	M 1.02	U
	OR		
2	Describe the operation of signals with amplitude.	M 1.04	U
3	Explain the classification of systems based on memory.	M 2.03	U
	OR		
4	Explain any 3properties of systems.	M 2.04	U
5	Describe the properties of Fourier series.	M 3.02	U
	OR		
6	Explain discrete time Fourier series.	M 3.01	U
7	Explain the properties of Laplace transform.	M 4.03	U
	OR		
8	Find the inverse Laplace transform of	M 4.04	A

	$$\frac{2}{s(s+2)(s-3)}$$		
9	Explain the classification of signals.	M 1.03	U
	OR		
10	Explain the time based operation of signals.	M 1.04	U
11	Explain the methods by which aliasing can be minimized.	M 3.03	U
	OR		
12	Explain the Fourier transform of continuous time non-periodic signal.	M 3.04	U

TED (21) – 2022 REVISION 2021

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

Model Question Paper II

SIGNALS AND SYSTEMS

Time: 3 Hour *Maximum Marks: 75*

PART A

I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

(9 * 1=9 marks)

		Module outcome	Cognitive level
1	What is continuous time signal?	M 1.03	R
2	List any two elementary signal.	M 1.02	R
3	Draw a ramp signal.	M 1.02	R
4	Define a static system.	M 2.03	R
5	What is time invariant system?	M 2.04	R
6	What do you mean by aliasing?	M 3.03	R
7	List any two properties of Fourier series.	M 3.02	R
8	Define Laplace transform.	M 4.01	R
9	Write the Laplace transform for step input.	M 4.01	U

PART B

II. Answer any **eight** questions from the following, each question carries 3 marks.

(8 * 3=24 marks)

		Module outcome	Cognitive level
1	Distinguish continuous time and discrete time signals.	M 1.03	U
2	Illustrate the time reversal of signals with figure.	M 1.04	U
3	Write notes on exponential function.	M 1.02	U

4	Differentiate stable and unstable system.	M 2.04	U
5	Define a discrete time system with example.	M 2.02	R
6	What are the properties of Fourier series?	M 3.02	R
7	What do you mean by reconstruction of signals?	M 3.03	R
8	Differentiate Fourier series and Fourier transform?	M 3.04	U
9	Find the Laplace transform of unit impulse function.	M 4.01	U
10	Find the inverse Laplace transform of $\frac{1}{s^2}$	M 4.04	A

PART C

III. Answer ALL questions. Each question carries 7 marks.

(6*7=42 marks)

		Module outcome	Cognitive level
1	Explain the classification of signals.	M 1.03	U
	OR		
2	Describe the basic operations of signals with time.	M 1.04	U
3	Distinguish between linear and nonlinear systems with examples.	M 2.03	U
	OR		
4	Define a system and explain how it is represented in its time domain.	M 2.01	U
5	Explain sampling theorem. What is aliasing?	M 3.03	U
	OR		
6	Explain Fourier transform of continuous time non periodic signal.	M 3.04	U
7	Find the Laplace transform of $$f(t) = e^{at} \sin \omega t$$	M 4.01	U

	OR		
8	Explain the differentiation theorem and integration theorem of Laplace transform.	M 4.03	U
9	Define signals and represent various types of signals with figures.	M 1.02	U
	OR		
10	Explain the arithmetic operations onsignals.	M 1.04	U
11	Derive the Laplace transform of a) Parabolic function b) Sine function c) Exponential function	M 4.01	U
	OR		
12	Find the inverse Laplace transform of $\frac{1}{s(s^2+5s+6)}$	M 4.04	A